

Lab - .NET - List Blobs

To Begin with the Lab

Summary

In this lab, you learn how to retrieve and display all blobs (files) stored inside an Azure Blob Storage Container using a .NET application. The application connects to the Azure Storage Account, accesses the Blob Container, and lists each blob by iterating through the collection of blobs. This functionality is commonly used to display uploaded files, documents, images, and backups stored in Azure Blob Storage.

Understanding

Azure Blob Storage stores files as **Blobs** inside a **Blob Container**. To view the files stored in a container, the application first connects to the Azure Storage Account using a **Storage Account Connection String**. It then creates a **BlobContainerClient** to access the desired container.

The `GetBlobsAsync()` method retrieves all blobs from the container. Using the `await foreach` loop, the application iterates through each blob and displays its name. This method is efficient because it retrieves blob information asynchronously, making it suitable for containers containing a large number of files.

Example

Suppose your Blob Container contains the following files:

Blob Container:

images

Stored Blobs:

01.py

profile.jpg

document.pdf

logo.png

Application Flow:

ASP.NET/.NET Application



Storage Account



Blob Container (images)



`GetBlobsAsync()`



01.py

profile.jpg

document.pdf

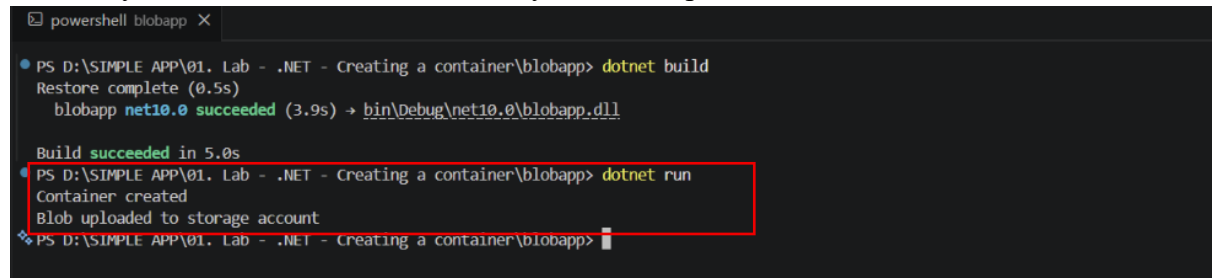
logo.png

The application displays the names of all blobs stored inside the **images** container.

Lab Steps

Step 1: Open the .NET Project

- Open your .NET project in Visual Studio or Visual Studio Code.
- Ensure the Azure Blob Storage package is already installed.
- Verify that the Blob Container already contains uploaded files.



```
powershell blobapp X

PS D:\SIMPLE APP\01. Lab - .NET - Creating a container/blobapp> dotnet build
Restore complete (0.5s)
blobapp net10.0 succeeded (3.9s) -> bin\Debug\net10.0/blobapp.dll

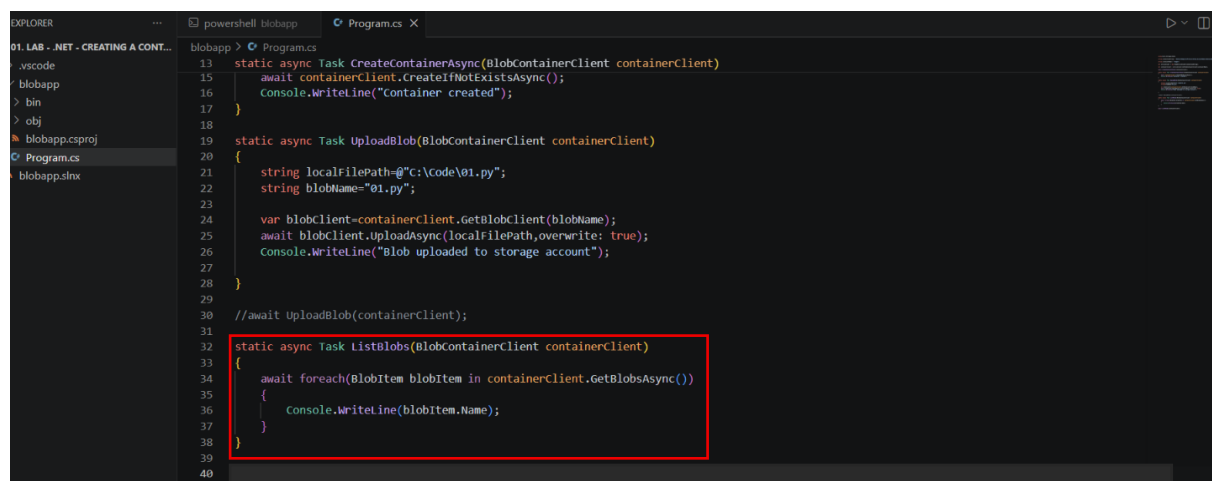
Build succeeded in 5.0s
PS D:\SIMPLE APP\01. Lab - .NET - Creating a container/blobapp> dotnet run
Container created
Blob uploaded to storage account
PS D:\SIMPLE APP\01. Lab - .NET - Creating a container/blobapp>
```

Step 2: Create the ListBlobs Method

- Create a method to **retrieve all blobs** from the container.

```
static async Task ListBlobs(BlobContainerClient containerClient)
{
    await foreach (BlobItem blobItem in containerClient.GetBlobsAsync())
    {
        Console.WriteLine(blobItem.Name);
    }
}
```

- Calls GetBlobsAsync() to **retrieve all blobs**.
- Uses await foreach to iterate through each blob.
- Prints the **blob name to the console**.



```
EXPLORER 01. LAB - .NET - CREATING A CONT...
  .vscode
  blobapp
  > bin
  > obj
  blobapp.csproj
  Program.cs
  blobapp.slnx

blobapp > Program.cs
13 static async Task CreateContainerAsync(BlobContainerClient containerClient)
14 {
15     await containerClient.CreateIfNotExistsAsync();
16     Console.WriteLine("Container created");
17 }
18
19 static async Task UploadBlob(BlobContainerClient containerClient)
20 {
21     string localFilePath=@"C:\code\01.py";
22     string blobName="01.py";
23
24     var blobClient=containerClient.GetBlobClient(blobName);
25     await blobClient.UploadAsync(localFilePath,overwrite: true);
26     Console.WriteLine("Blob uploaded to storage account");
27 }
28
29 //await UploadBlob(containerClient);
30
31
32 static async Task ListBlobs(BlobContainerClient containerClient)
33 {
34     await foreach (BlobItem blobItem in containerClient.GetBlobsAsync())
35     {
36         Console.WriteLine(blobItem.Name);
37     }
38 }
39
40
```

Step 3: Call the Method

Call the method after creating the BlobContainerClient.

```
await ListBlobs(containerClient);
```

- Executes the blob listing operation.

- **Displays all blob names** stored in the container.

```
static async Task ListBlobs(BlobContainerClient containerClient)
{
    await foreach(BlobItem blobItem in containerClient.GetBlobsAsync())
    {
        Console.WriteLine(blobItem.Name);
    }
}

await ListBlobs(containerClient);
```

Step 4: Build the Project

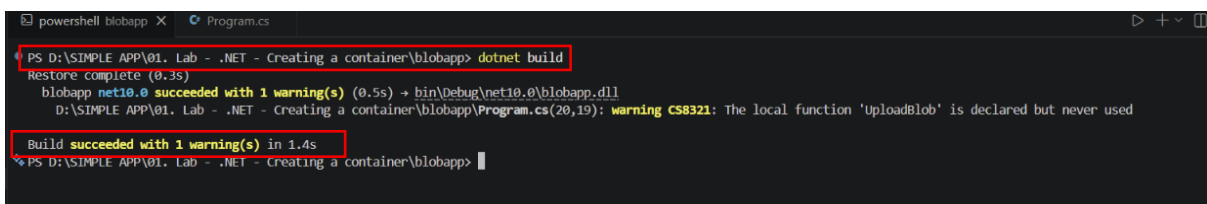
Run the following command:

```
dotnet build
```

- **Compiles** the application.
- Verifies that there are no build errors.

Expected Output:

Build succeeded.



```
PS D:\SIMPLE APP\01. Lab - .NET - Creating a container\blobapp> dotnet build
Restore complete (0.3s)
blobapp net10.0 succeeded with 1 warning(s) (0.5s) -> bin\Debug\net10.0\blobapp.dll
D:\SIMPLE APP\01. Lab - .NET - Creating a container\blobapp\Program.cs(20,19): warning CS8321: The local function 'UploadBlob' is declared but never used
Build succeeded with 1 warning(s) in 1.4s
PS D:\SIMPLE APP\01. Lab - .NET - Creating a container\blobapp>
```

Step 5: Run the Application

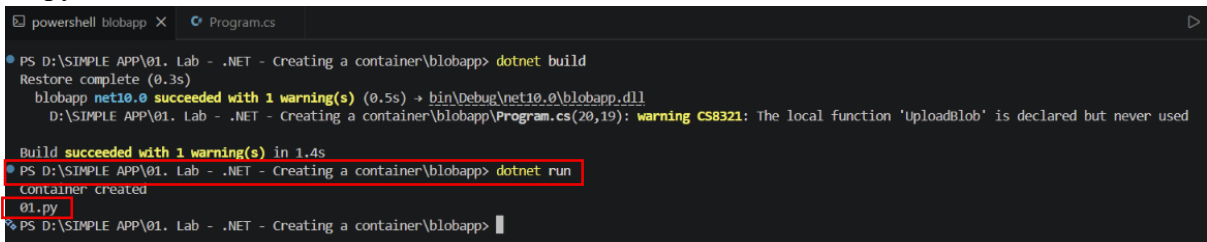
Run the following command:

```
dotnet run
```

- Starts the .NET application.
- Retrieves all blobs from the Azure Blob Container.
- Displays their names in the terminal.

Example Output:

01.py



```
PS D:\SIMPLE APP\01. Lab - .NET - Creating a container\blobapp> dotnet build
Restore complete (0.3s)
blobapp net10.0 succeeded with 1 warning(s) (0.5s) -> bin\Debug\net10.0\blobapp.dll
D:\SIMPLE APP\01. Lab - .NET - Creating a container\blobapp\Program.cs(20,19): warning CS8321: The local function 'UploadBlob' is declared but never used
Build succeeded with 1 warning(s) in 1.4s
PS D:\SIMPLE APP\01. Lab - .NET - Creating a container\blobapp> dotnet run
Container created
01.py
PS D:\SIMPLE APP\01. Lab - .NET - Creating a container\blobapp>
```